



Figure 3: LLM-Prop performance per (a) label scaling strategy and (b) input sequence length. For the band gap and volume, the lower the better. While for the Is-gap-direct, the higher the better. The performance of each model is calculated on validation set.

with its original tokenizer and without preprocessing the input (LLM-Prop-baseline) and then add to it each strategy separately. We also compare the performance of each strategy to when all strategies are combined together to make LLM-Prop (LLM-Prop+all). Overall, LLM-Prop+all provides the best results except on volume prediction where it shows comparable performance with the LLM-Prop+[CLS] version. Among all strategies, adding the [CLS] token for pooling seems to give the best improvement on all property prediction tasks. Replacing bond distances with the [NUM] token tends to slightly harm the performance, especially on volume prediction, compared to replacing bond angles with [ANG]. Label scaling significantly improves performance on both band gap and volume prediction while removing stopwords slightly improves the performance on band gap and volume prediction, but slightly harms the performance on predicting whether the band gap is direct or indirect. The modified tokenizer slightly improves the performance of LLM-Prop on band gap prediction but harms the performance on predicting other properties.

How does LLM-Prop perform with respect to the label scaling strategy? We also analyze how each label scaling technique impacts performance on regression tasks. The results in Figure 3(a) show that when predicting band gap the performance difference among label normalization strategies is not significant while for volume prediction, z-score normalization (z_norm) significantly outperforms other normalization schemes.

How does LLM-Prop perform with respect to the number of input tokens? Figure 3(b) shows the performance of LLM-Prop as a function of the input sequence length. The results show that there is a clear correlation between performance and input sequence length. Accounting for longer sequences tends to improve performance. Therefore, by default, LLM-Prop is set to process up to 888 input tokens which is the maximum length that could be processed by the NVIDIA RTX A6000 GPUs that we used for training. And, as the results show, we believe that accounting for longer input sequence length will yield even better performance on crystal property prediction.